Lorin Urbantat

MSc Computer Science, ETH Zurich · Machine Learning & Interactive Systems

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ETH computer science master's student with experience in Unity 3D, AI applications, and speech processing. Interested in conditional generative models for expressive avatar motion from sparse head, inertial, and speech signals.

EDUCATION

ETH Zurich, MSc Computer Science 2025 - 2027
- Expected Graduation: September 2027
- Major: Data Management Systems; Minor: Machine Learning
- Relevant coursework: Machine Perception, Computational Intelligence Lab, Computational Semantics for Natural Language Processing (5.75/6)

ETH Zurich, BSc Computer Science 2021 - 2024
- Graduated August 2024, GPA 5.04 (6.0 – highest)
- Visual Computing (5.75/6), Numerical Methods (5.75/6), Autonomous Mobile Robots (5.25/6), Human Computer Interaction (5.25/6), Introduction to Machine Learning

WORK EXPERIENCE

Information Security Lab Teaching Assistant, ETH Zurich (part-time) 2026 - Present
- Test lab exercises and host exercise sessions

Software Engineer, ETH Chair of Education Systems (part-time) 2025 - Present
- Sole engineer for a research-focused RAG chatbot serving 1,000 users. Built the application and document processing pipeline, including OCR and image descriptions, using Next.js, Convex and Microsoft Foundry.

Software Engineer, ETH Game Technology Center (part-time) 2023 - 2025
- Morph Tales Game, Unity 3D
- Developed features for an interactive Unity 3D game and worked on generative AI in game characters.
- Built a game analytics pipeline using Firebase, BigQuery and Looker Studio.

Coach, mint & pepper (part-time) 2025
- STEM and AI tutor for high school students

Calculus Teaching Assistant, ETH Zurich (part-time) 2023
- Host exercise classes, 15 - 30 students

SELECTED PROJECTS & RESEARCH EXPERIENCE

TLDR - Speech-based Lecture Summarizer, VIScon Hackathon 2023
- Used OpenAI Whisper and language model APIs to turn lecture recordings into searchable summaries.

SCION Control Plane Algorithms Test Environment, Bachelor Thesis 2024
- Built a Docker-based test environment and traffic generator to evaluate algorithms under configurable network topologies and changing link conditions.

Unikraft on Raspberry Pi 5, ETH Semester Project 2026
- Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers; developed a cache-safe kernel reload mechanism with OpenOCD and GDB.

Flateroo, Flat Sharing Platform 2024 - 2025
- Built a React Native application with real-time chat and video streaming; implemented video transcoding, HLS delivery and a PostgreSQL backend.

SKILLS & INTERESTS

Languages	German (native), English (fluent), French (DELF B1)
Technologies	Python, PyTorch, C, C++, Unity 3D, Machine Learning, RAG, SQL, TypeScript, React, Git, Linux, Docker
Interests	Biking, Sports, Cooking, Dancing, Singing